New Eden

Steel-Frame Implementation Plan (Autovol-Style Automation)

How New Eden's planned physical factory could adopt an Autovol-style automated framing-table system — the ABB-robot-plus-systems-integrator model used by Autovol and House of Design Robotics for timber — fully re-engineered for cold-formed steel. This is a planning document only; it makes no compliance claims and tracks no dollar budget. Companies and products cited are real and linked to primary sources; where a claim could not be independently verified, that is stated explicitly rather than assumed.

Executive Summary

Autovol (Nampa, Idaho) proves the model New Eden should copy structurally, not just visually: buy stock ABB industrial robots, then pay a dedicated systems integrator — House of Design Robotics, an ABB-recognised robotics integrator/value provider founded 2012 in Nampa — to supply everything ABB doesn't: grippers, end-of-arm tooling, gantries, jig tables, conveyors and the CAD-to-robot programming layer. Autovol's own framing is timber (dimensional lumber, OSB, pneumatic nail-gangs) on a Güdel-gantry-based line. New Eden is steel-only, so the tooling — not the robot brand — has to change: screw-gang or resistance-spot-weld heads instead of nail guns, magnetic/mechanical grippers instead of vacuum cups for thin open profiles, and framing tables sized and jigged for cold-formed C-stud/track sections rather than sawn lumber. The closest real-world precedent for a robotic steel version of this already exists and is operating: Randek AB's ZeroLabor robotic system, installed at British Offsite's Horizon factory in Braintree, UK — eight robots running a robotic light-gauge-steel panel assembly line. That is the single strongest proof point that an 'Autovol for steel' is buildable today, not speculative.

1. Vendor Landscape — Real Automated Cold-Formed Steel Framing Systems

Every vendor below was checked against its own published pages at the time of writing (July 2026). Confirmed-real, confirmed-product entries are marked accordingly; unverifiable claims are flagged.

Vendor	Status	What they actually sell
Randek AB (Sweden)	Confirmed real. Closest Autovol analogue for steel.	Wall/floor/roof production lines; a dedicated **Light Gauge Steel Frame System**; the **AutoWall S3000** semi-automated wall line; and **ZeroLabor**, a fully robotic system. British Offsite's £45M Horizon factory (Braintree, UK) runs 8 Randek-supplied robots in three teams performing zero-labour cells — insulation fitting, plasterboard fitting, window-opening cutting — on light-gauge steel panels. This is the first robotic LGS panel assembly line in the UK and the strongest real-world precedent for this whole report.
FRAMECAD (NZ / global)	Confirmed real.	End-to-end roll-forming + CAD/CAM: machines (e.g. F325iT, F450iT) roll-form coil into studs/track and punch/notch/dimple per the digital model in one pass, with an Auto-Gauging System and stated <0.01mm punch/cut tolerance. Strong on roll-forming-to-CAM; no confirmed dedicated robotic fastening-gang/framing-table product distinct from the roll former itself.
Howick Ltd (Auckland, NZ)	Confirmed real.	Precision roll-forming machines converting digital design files directly into steel wall, floor, truss and panel components; stated ±0.5mm accuracy; machines in 80+ countries; proprietary telescopic-framing and load-bearing-joint techniques. Roll-forming/CAM focus, similar category to FRAMECAD; not a robotic assembly-table vendor per se.

MiTek (Inc. / APAC / South Africa)	Confirmed real; steel product line confirmed, automated-steel-table claim unverified.	Strong automated jig-table heritage (e.g. Wizard PDS perimeter-definition jig system, laser-projected layout, flipper-arm truss tables) is documented for **timber** truss/wall tables. MiTek also sells cold-formed light-gauge steel systems (Aegis / Ultra-Span, SteelEngine) as design software and connectors. No public source found confirming a MiTek-branded automated steel framing table equivalent to its timber Multitruss-class lines — treat as unconfirmed, not as a candidate deliverable, until verified directly with MiTek.
voestalpine Metsec (UK)	Confirmed real; not an automation-equipment vendor.	Manufactures SFS (Steel Framing System) lipped C-section studs/track and connection components, supplied for site or factory assembly. Metsec is a materials/component supplier, not a framing-table or robotics vendor — relevant to New Eden as a possible coil/section source or profile reference, not as an automation partner.
Voortman Steel Machinery (NL)	Confirmed real; wrong market segment.	Fully automated CNC beam/plate/angle processing with Multi System Integration (MSI) software and an automatic fitting-and-welding line — but for structural steel fabrication (beams, connections), not light-gauge residential framing. Adjacent technology, not a direct fit.
"Bright Steel"	Could not verify.	No company matching this name and offering automated steel framing tables was found in search. Do not cite or rely on this name until a source can independently confirm it; treat as an open research item, not a vendor.

Sources:

Randek — ZeroLabor robotic system

Randek — AutoWall S3000 system

Randek — Light Gauge Steel Frame System

Light Steel Frame Association — Randek/British Offsite robotic LGS line

Built Offsite — British Offsite's Horizon factory, 8 Randek robots

FRAMECAD — Steel Framing Machines

FRAMECAD — F450iT

Howick Ltd — homepage

Howick Ltd — Why Howick (accuracy, country count)

MiTek Inc. — Automated Solutions

MiTek US — Machinery

voestalpine Metsec — Steel Framing SFS

Voortman Steel Machinery — Multi System Integration

2. Robot-Plus-Integrator Model, Not ABB-Alone

Autovol's own framing: "you can buy an ABB robot but you're going to need a House Of Design Robotics company to provide all the value-added engineering services." House of Design Robotics (Nampa, Idaho, founded 2012) is documented as an ABB Robotics integrator/value provider; it supplies the grippers, end-of-arm tooling, Güdel-brand overhead gantries, jig tables, conveyors and the CAD-to-robot programming software that turn a bare ABB arm into a working framing line. ABB itself confirms the robot count: Autovol's 400,000 sq ft plant runs 53 ABB six-axis robots automating wall, ceiling and floor production.

New Eden's vision doc already commits to ABB robots as the robotics backbone, and the digital twin already models an ABB IRB 8700-550/4.2 for the truss-welding cell. That is a sound anchor — ABB is real, its specs are public, and it is a legitimate industrial choice — but the vision doc should not assume the robot arm alone is a drop-in framing solution. The IRB 8700 is a 550 kg-payload, 4.2 m-reach heavy robot suited to structural welding, not to picking light 0.55–1.6 mm gauge steel studs off a magazine at speed; a framing-table cell would likely use a smaller/faster ABB class (e.g. the mid-payload articulated families used for pick-and-place and material handling) alongside the IRB 8700 welding cell, not instead of it.

What a systems-integration partner needs to deliver for steel (vs. Autovol's timber tooling):

Function	Autovol (timber)	New Eden (steel) requirement
End-of-arm gripper	Vacuum-cup sheet lifters for OSB/lumber panels.	Vacuum cups don't grip open C-channel/track profiles reliably. Needs mechanical/magnetic or internal-expansion-jaw grippers shaped to the stud/track flange and web, sized per profile depth (e.g. 64/92/150 mm C-studs), likely a tool-change rack for different member types.
Fastening head	Pneumatic nail-gang head on the Floor Gantry.	No nails in steel. Needs a multi-spindle self-drilling screw-gang head (screw fastening is the industry-standard cold-formed connection per AISI/CFSEI guidance) and/or a resistance spot-weld gang for higher-volume factory joints, since resistance welding is documented as faster and more automation-friendly than screwing for thin-to-thin steel joins.
Gantry / table sizing	Sized for lumber stud spans and OSB sheet dimensions.	Steel members are dimensionally similar in length but the table/jig needs squaring stops and clamps rated for steel's higher stiffness and tighter dimensional tolerance, plus a different clamping force profile so thin-gauge (as low as ~0.55 mm) webs aren't deformed by clamps sized for lumber.
CAD-to-robot software	Programs robots directly from the building's timber 3D model.	Same concept, different profile library — the software must be re-taught cold-formed steel C-stud/track geometry (AISI S240 profile families) instead of dimensional lumber sizes, and needs to output screw/weld point patterns instead of nail patterns.

Güdel is a confirmed real linear-motion/gantry manufacturer (Güdel Inc., part of the Güdel Group) with published overhead TrackMotion (TMO) and multi-axis gantry products, and third-party video documents a Güdel gantry being installed at House of Design's facility — consistent with the companion research already on file. Randek's own robot arms at British Offsite are described only as "Randek Robotics"-supplied; no source found confirms which underlying robot brand (ABB, KUKA, or an in-house arm) they are built on — flagged as an open item, not assumed to be ABB.

3. Where This Slots Into New Eden's Production Flow

From the Master Vision & System Architecture document, the canonical flow is:

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Steel Coil Arrival
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Automated Storage
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Roll Forming
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Robotic Pick and Place
↓
Wall Frame Assembly
↓
Roof Truss Assembly
↓
Floor Frame Assembly
↓
Conveyor Transfer
↓
Module Assembly
↓
Electrical Installation
↓
Plumbing Installation
↓
Insulation
↓
Windows
↓
Doors
↓
Internal Linings
↓
Exterior Cladding
↓
Painting / Finishing
↓
Quality Assurance
↓
Final Inspection
↓
Dispatch
```

An Autovol-style framing-table system is not a new stage bolted onto this flow — it is the concrete automation layer that **replaces the manual-labour version of three consecutive stages**: **Wall Frame Assembly, Roof Truss Assembly, and Floor Frame Assembly**. Each becomes its own robotic framing-table cell (mirroring the digital twin's existing three parallel Walls / Roof / Floors lanes converging at a Marriage Station), fed by **Roll Forming** and **Robotic Pick and Place** upstream and handing finished panels/trusses/floor cassettes to **Conveyor Transfer** and then **Module Assembly** downstream.
Interfaces this system must satisfy:
Upstream — Roll Forming output: the framing table's infeed must accept cut-to-length C-stud and C-track sections directly off the roll former (or off automated storage buffering roll-formed stock), pre-punched with any connection/service holes the CAD-to-robot software already specified at roll-forming time — this is where FRAMECAD/Howick-style auto-gauging and punch/notch capability earns its keep, since it removes a manual measuring/marking step before framing even starts.

Downstream — Conveyor Transfer / Module Assembly: finished wall, roof and floor sub-assemblies must exit the framing tables onto self-propelled transport carts or powered conveyor (Autovol's AUTO-ROLL-class conveyor is the reference category here, though the specific product name could not be independently re-verified from public sources in this research pass — treat the name as reported context, not confirmed) sized and squared for delivery to the Marriage/Module Assembly station where walls, roof and floor meet the six-sided module shell.

4. Station-by-Station Equipment List

Mapped against the vision doc's own list of "potential robotic tasks" (loading roll formers, transferring steel sections, welding, fastening, screwing, adhesive application, lifting wall frames, rotating assemblies, positioning roof sections, CNC material handling, quality inspection, palletising, packaging).

Station	Robotic task (vision doc)	Equipment
Roll Forming infeed	Loading roll formers	Coil decoiler + robotic/automated coil loading; FRAMECAD- or Howick-class roll former with in-line CAD/CAM punch-cut-notch control.
Roll Forming outfeed / Robotic Pick and Place	Transferring steel sections	ABB mid-payload articulated arm (or gantry-mounted picker) lifting cut studs/track onto sorting racks or directly onto the next framing table's magazine.
Wall Frame Assembly	Fastening, screwing, lifting wall frames, rotating assemblies	Steel jig/framing table (squaring fences, clamps) + overhead gantry (Güdel-class TrackMotion/gantry) carrying an ABB arm with quick-change gripper; multi-spindle screw-gang head and/or resistance spot-weld gang; panel flip/rotate fixture for two-sided fastening.
Roof Truss Assembly	Positioning roof sections, welding, fastening	Truss jig table with positioning pucks/lasers (MiTek-style jig concept, re-tooled for steel); ABB IRB 8700-class heavy robot for structural welds at chord/web connections; screw-gang for lighter joints.
Floor Frame Assembly	Fastening, screwing, CNC material handling	Floor-cassette framing table sized for joist spans; screw-gang head; robotic handling of any embedded service penetrations (CNC-cut per digital model).
Between framing cells / QA	Quality inspection	Laser measurement/scanning arm or fixed laser QA gate checking squareness, member spacing and screw/weld point presence against the CAD model before release to Conveyor Transfer.
Conveyor Transfer	Transferring steel sections	Powered conveyor and/or self-propelled transport carts moving finished wall/roof/floor sub-assemblies to Module Assembly (Marriage Station).
Module Assembly / Dispatch	Palletising, packaging	Robotic palletising/packaging arm for hardware kits, offcuts and ancillary components; not applicable to whole modules, which move by crane/transporter rather than palletiser.

5. Key Technical Deltas vs. Autovol's Timber Process

Factor	Autovol / timber	New Eden / cold-formed steel
Fastening method	Pneumatic nail-gang (nail plates for trusses).	Self-drilling screws (industry-standard per CFSEI/AISI guidance) for field assembly; resistance spot welding as a faster, more automation-friendly option for thin-to-thin factory joints — not conventional arc welding, which is generally unsuited to sheet as thin as 0.4-1.0 mm.
End effector	Vacuum cups grip solid sheet goods (OSB, lumber) well.	Open C-profiles have little flat surface for vacuum; grippers need to clamp the flange/web geometry mechanically or magnetically and must be swappable per profile size.
Panel weight/rigidity	Lumber framing is heavier per member but more forgiving of clamp pressure and minor misalignment; OSB sheathing adds significant dead weight.	Steel members are lighter per linear metre but far less tolerant of localised clamp/gripper point loads on thin webs — over-clamping can permanently deform a 0.55 mm stud. Assembled steel panels are generally lighter overall, which changes lifting/rotating fixture load cases.
Tolerances	Dimensional lumber varies by species/moisture; framing tables compensate with generous tolerance bands.	Roll-formed steel holds tight, repeatable tolerances at the mill (Howick states ±0.5 mm; FRAMECAD states <0.01 mm at the punch/cut head) — the framing table and fastening heads should be designed to preserve that precision rather than the wider tolerance band timber framing tolerates.
Governing standard	Timber framing codes (not applicable to New Eden).	AISI S240, North American Standard for Cold-Formed Steel Structural Framing, governs the structural members and connections a real steel framing line would produce — cited here by name only, as the standard a real build would be engineered with reference to, not as a compliance claim.

6. Open Questions & Recommended Next Steps

Genuinely uncertain / not resolved by this research pass:

- Whether FRAMECAD or Howick (roll-forming/CAM specialists) or Randek (framing-line/robotics specialist) is the better lead vendor conversation for New Eden — they occupy different parts of the stack and a real programme likely needs more than one.
- What robot brand underlies Randek's ZeroLabor arms at British Offsite — not confirmed as ABB from public sources, which matters if New Eden wants to stay single-vendor on robots.
- Whether MiTek has an automated steel-specific framing table product beyond its documented timber jig-table lineage — needs a direct vendor enquiry, not assumed from public marketing pages.
- "Bright Steel" could not be identified as a real company — drop it as a candidate or ask the source of that name for a corrected reference.
- Exact clamp-force and gripper-geometry requirements for New Eden's specific stud gauges/profiles — this is an engineering exercise for a systems integrator, not something desk research can answer.

Recommended next steps:

- Prototype first: a single Wall Frame Assembly cell (one framing table, one gantry-mounted ABB arm, one screw-gang head) sized for one wall-panel family, before committing to Roof and Floor cells — this mirrors how Autovol's own line and Randek's ZeroLabor system are built up cell-by-cell.
- Contact Randek AB directly (via randek.com) to ask about a light-gauge-steel ZeroLabor reference configuration and lead times/capacity figures, since they are the only vendor found with an operating robotic steel-framing line in production.

- Contact FRAMECAD and Howick to compare their auto-gauging/CAM roll-forming lines as the upstream feed for whichever framing-table cell is built first — both are real, mature, and already export internationally.
- Contact House of Design Robotics directly to ask whether their systems-integration scope (grippers, gantries, CAD-to-robot software) could be re-engineered for steel, given they already do this exact integration role for Autovol's timber line and are an established ABB integrator.
- Treat resistance spot-welding vs. self-drilling screw fastening as an open engineering decision, not a foregone conclusion — get a structural engineer's opinion on which connection type suits New Eden's specific stud gauges and load paths before tooling a fastening head.

Verification Notes

Confirmed real via primary/company sources in this research pass: FRAMECAD, Howick Ltd, MiTek, voestalpine Metsec, Voortman Steel Machinery, Randek AB, British Offsite, Güdel, Autovol, House of Design Robotics, ABB. Not confirmed / flagged as open: "Bright Steel" as a company name; the specific product name "AUTO-ROLL" for Autovol's conveyor (treated as reported context, not independently re-verified); the robot brand underlying Randek's ZeroLabor arms; a MiTek automated steel-specific (as opposed to timber) framing-table product.

Compliance & Budget Disclaimer

This document is a planning and research reference. It does not claim, and cannot claim, that any described configuration is "AS 4100 compliant," "AS/NZS 4600 compliant," "NCC compliant," or certified against any standard — standards are cited by name/number only, as references a real engineering process would need to work against. This document does not track, estimate, or commit to any dollar budget for New Eden.